

# Energy transport in plane waves

Paul Haskins

March 28, 2025

Poynting vector :

$$\vec{s} = \vec{E} \times \vec{H} \quad (1)$$

if we assume linear media

$$= \frac{1}{\mu} \vec{E} \times \vec{B} \quad (2)$$

Energy density

$$u = \frac{1}{2} (\vec{H} \cdot \vec{B} + \vec{E} \cdot \vec{D}) = \frac{1}{2} \left( \frac{1}{\mu} \vec{B}^2 + \epsilon \vec{E}^2 \right) \quad (3)$$

quadratic in the physical fields.

To correctly compute  $\vec{S}, u$  one computes

$$\vec{S} = \frac{1}{\mu} \Re \vec{E} \times \Re \vec{B} \neq \frac{1}{\mu} \Re (\vec{E} \times \vec{B}) \quad (4)$$

if  $\vec{E}, \vec{B}$  and our complex fields

$\exists$  a shortcut via time averaging, visible light  $\lambda = \mathcal{O}(100nm)$ , in a vacuum  $\lambda v = c$ ,  $v = \frac{1}{T}$  with  $T \equiv$  period

$$T = \frac{\lambda}{c} \approx \frac{10^{-7}m}{3 \cdot 10^8 \frac{m}{s}} = \frac{1}{3} \times 10^{-15}s \quad (5)$$

$\rightarrow$  shorter than the time resolution of most experiments.

observation

$$\begin{cases} a(t) = a_0 e^{-i\omega t} \\ b(t) = b_0 e^{-\omega t} \end{cases} \quad \text{otherwise} \quad (6)$$

$$(\Re a(t))(\Re b(t)) = \frac{1}{2} (a(t) + a^*(t)) \frac{1}{2} (b(t) + b^*(t)) \quad (7)$$

$$= \frac{1}{4} (a_0 b_0 e^{-2i\omega t} + a_0 b_0^* + a_0^* b_0 + a_0^* b_0^* e^{2i\omega t}) \quad (8)$$

We introduce the time average

$$\bar{f}(t) = \frac{1}{T} \int_t^{t+T} f(t') dt' \quad (9)$$

$$\frac{1}{T} \int_t^{t+T} dt' e^{\pm 2i\omega t'} = \frac{1}{T} \frac{1}{\pm 2i\omega} e^{\pm 2i\omega t'} \Big|_t^{t+T} \quad (10)$$

$$\implies T = \frac{1}{\nu} = \frac{2\pi}{\omega} \quad (11)$$

$$\overline{(\Re a(t))(\Re b(t))} = \frac{1}{4} (a_0 b_0^* + a_0^* b_0) = \frac{1}{2} (a_0 b_0^*) = \frac{1}{2} \Re (a(t) b^*(t)) \quad (12)$$

Rather than taking real parts of both we can take the conjugate of the second then cross them and multiply by a half. something along those lines.

$$\vec{s} = \frac{1}{2} \Re (\vec{E} \times \vec{H}^*) \quad (13)$$

$$\text{linear media} = \frac{1}{2\mu} \Re (\vec{E} \times \vec{B}^*) \quad (14)$$

$$\bar{u} = \frac{1}{4} \Re (H \cdot B^* + \vec{E} \cdot \vec{D}^*) = \frac{1}{4} \Re \left( \frac{1}{\mu} \vec{B} \cdot \vec{B}^* + \epsilon \vec{E} \cdot \vec{E}^* \right) \quad (15)$$

$$\vec{\nabla} = |\vec{B}|^2 = (\Re \vec{B})^2 + (\Im \vec{B})^2 \neq \Re |\vec{B}|^2 \quad (16)$$

Poynting vector

$$\vec{s} = \frac{1}{2\mu} \Re (\vec{E} \times \vec{B}^*) = \frac{1}{2\mu\omega} [\vec{E}_0 \times (\vec{k} \times \vec{E}_0)] \quad (17)$$

$$= \frac{1}{2\omega\mu} \Re (\vec{k} \vec{E}_0 \vec{E}_0^* - \vec{E}_0^* (\vec{E}_0 \cdot \vec{k})) = \frac{1}{2} \frac{k}{\omega} \frac{1}{\mu} \vec{E}_0 \vec{E}_0^* \cdot \vec{k} \quad (18)$$

$$= \frac{k}{\omega} \frac{1}{\mu\epsilon} \vec{u} \hat{k} = v \vec{u} \hat{k} \quad (19)$$

where  $\frac{k}{\omega} = \frac{1}{v}$  and  $\frac{1}{\mu\epsilon} = v^2$

$\vec{s}$  is an energy current density, energy density time velocity. units

$$\left[ \frac{\vec{s}}{s} \right] = \frac{\text{energy}}{3 \text{ volume}} \cdot \text{velocity} = \frac{\text{energy}}{\text{area} \cdot \text{time}} = \frac{\text{power}}{\text{area}} = \frac{W}{m^2} \quad (20)$$

Radiation power

$$\frac{\vec{s}}{v} = \frac{\text{force}}{\text{area}} \quad (21)$$

Typical question calculate the time energy exactly or time average

(Reflection and transmission of EM waves at interfaces between linear dielectric media e.g water, air, glass ...)

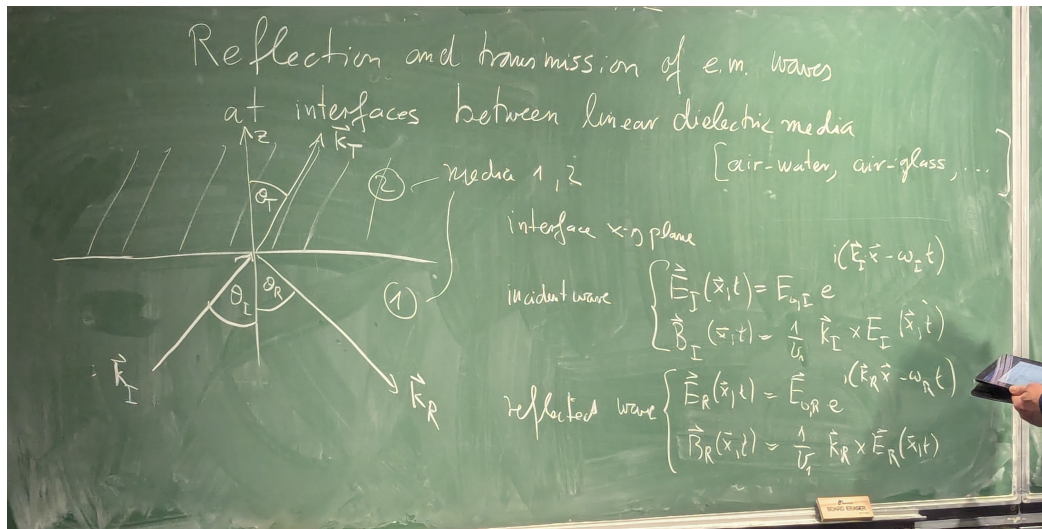


Figure 1: 8.1.jpg

While we have the transmitted wave to be

$$\begin{cases} \vec{E}_T(\vec{x}, t) = \vec{E}_{0T} e^{i(\vec{k}_T \cdot \vec{x} - \omega_T t)} \\ \vec{B}_T(\vec{x}, t) = \frac{1}{v_2} \vec{k}_T \times \vec{E}_T \end{cases} \quad (22)$$

questions

- Relation between  $\vec{k}_I, \vec{k}_R, \vec{k}_T$
- Relation between  $\omega_I, \omega_R, \omega_T$
- size of reflected/transmitted amplitudes.
- same as the above one for intensities
- what about polarization?